Exercise 10.2 SOLUTION

Q1. Without using table find the values

i $\mathbf{Sin\ 15^\circ = Sin\ (45^\circ - 30^\circ)}$

Sol: $\sin 15^\circ = \text{Sin } 45^\circ \text{ Cos } 30^\circ - \text{Cos } 45^\circ \text{ Sin } 30^\circ$

$$= \left(\frac{1}{\sqrt{2}}\right)\left(\frac{\sqrt{3}}{2}\right) - \left(\frac{1}{\sqrt{2}}\right)\left(\frac{1}{2}\right) = \frac{\sqrt{3}}{2\sqrt{2}} - \frac{1}{2\sqrt{2}} = \frac{\sqrt{3}-1}{2\sqrt{2}}$$

ii. **Cos15°= Cos (45° – 30°)**

Sol: $\cos 15^\circ = \text{Cos}45^\circ \text{ Cos } 30^\circ + \text{Sin}45^\circ \text{ Sin}30^\circ$

$$= \left(\frac{1}{\sqrt{2}}\right)\left(\frac{\sqrt{3}}{2}\right) + \left(\frac{1}{\sqrt{2}}\right)\left(\frac{1}{2}\right) = \frac{\sqrt{3}}{2\sqrt{2}} + \frac{1}{2\sqrt{2}} = \frac{\sqrt{3}+1}{2\sqrt{2}}$$

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iii. **tan15°= tan (45° – 30°)**

Sol: $\tan 15^\circ = \dfrac{\tan 45^\circ - \tan 30^\circ}{1+\tan 45^\circ \tan 30^\circ} = \dfrac{1-\frac{1}{\sqrt{3}}}{1+1.\frac{1}{\sqrt{3}}} = \dfrac{\frac{\sqrt{3}-1}{\sqrt{3}}}{\frac{\sqrt{3}+1}{\sqrt{3}}} = \dfrac{\sqrt{3}-1}{\cancel{\sqrt{3}}} \times \dfrac{\cancel{\sqrt{3}}}{\sqrt{3}+1} = \dfrac{\sqrt{3}-1}{\sqrt{3}+1}$

iv. **Sin105°= Sin (60° + 45°)**

Sol: $\sin 105^\circ$ = Sin60° Cos45° + Cos60° Sin45°

$= (\frac{1}{\sqrt{2}})(\frac{\sqrt{3}}{2}) + (\frac{1}{\sqrt{2}})(\frac{1}{2}) = \dfrac{\sqrt{3}}{2\sqrt{2}} + \dfrac{1}{2\sqrt{2}} = \dfrac{\sqrt{3}+1}{2\sqrt{2}}$

v. **Cos105°= Cos (60° + 45°)**

Sol: $\cos 105^\circ$ = Cos60° Cos45° – Sin60° Sin45° $= \left(\frac{1}{2}\right)\left(\frac{1}{\sqrt{2}}\right) - \left(\frac{\sqrt{3}}{2}\right)\left(\frac{1}{\sqrt{2}}\right) = \dfrac{1}{2\sqrt{2}} - \dfrac{\sqrt{3}}{2\sqrt{2}} = \dfrac{1-\sqrt{3}}{2\sqrt{2}}$

vi. **tan105°= tan (60° + 45°)**

Sol: $\tan 105^\circ = \dfrac{\tan 60^\circ + \tan 45^\circ}{1-\tan 60^\circ \tan 45^\circ} = \dfrac{\sqrt{3}+1}{1-\sqrt{3}.1} = \dfrac{1+\sqrt{3}}{1-\sqrt{3}}$

Q2. Prove that

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i. **Sin (45°+ α) = $\dfrac{1}{\sqrt{2}}$ (Sinα + Cosα)**

Sol. L.H.S = Sin (45°+ α) = Sin 45° Cosα + Cos 45° Sinα = $\dfrac{1}{\sqrt{2}} Cos\alpha + \dfrac{1}{\sqrt{2}} Sin\alpha = \dfrac{1}{\sqrt{2}}(Sin\alpha + Cos\alpha)$

ii. **Cos (α + 45°) = $\dfrac{1}{\sqrt{2}}$ (Cosα – Sinα)**

Sol. L.H.S = Cos (α + 45°) = Cosα Cos 45° – Sinα Sin45°= Cosα $\dfrac{1}{\sqrt{2}} - Sin\alpha \dfrac{1}{\sqrt{2}} = \dfrac{1}{\sqrt{2}}$ (Cosα – Sinα)= R.H.S

Q3. Prove that

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i **tan (45° + A) tan (45° – A) = 1**

Sol. L.H.S = tan (45° + A). tan (45° – A)

$= \left(\dfrac{\tan 45^\circ + \tan A}{1-\tan 45^\circ \tan A}\right).\left(\dfrac{\tan 45^\circ - \tan A}{1+\tan 45^\circ \tan A}\right) = \left(\dfrac{1+\tan A}{1-1.\tan A}\right)\left(\dfrac{1-\tan A}{1+1.\tan A}\right)$

$= \left(\dfrac{\cancel{1+\tan A}}{\cancel{1-\tan A}}\right)\left(\dfrac{\cancel{1-\tan A}}{\cancel{1+\tan A}}\right) = 1 = R.H.S$

ii. $\tan\left(\dfrac{\pi}{4} - \theta\right) + \tan\left(\dfrac{3\pi}{4} + \theta\right) = 0$

Sol: L.H.S= $\tan\left(\dfrac{\pi}{4} - \theta\right) + \tan\left(\dfrac{3\pi}{4} + \theta\right) = \dfrac{\tan\frac{\pi}{4} - \tan\theta}{1+\tan\frac{\pi}{4}\tan\theta} + \dfrac{\tan\frac{3\pi}{4} + \tan\theta}{1-\tan\frac{3\pi}{4}\tan\theta} = \dfrac{1-\tan\theta}{1+1.\tan\theta} + \dfrac{-1+\tan\theta}{1-(-1)\tan\theta}$

$= \dfrac{1-\tan\theta}{1+\tan\theta} + \dfrac{-1+\tan\theta}{1+\tan\theta} = \dfrac{\cancel{1} - \cancel{\tan\theta} - \cancel{1} + \cancel{\tan\theta}}{1+\tan\theta} = \dfrac{0}{1+\tan\theta} = 0 = R.H.S$

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iii. $\mathbf{Sin\left(\theta+\frac{\pi}{6}\right)+Cos\left(\theta+\frac{\pi}{3}\right)=Cos\theta}$

Sol: L.H.S $= Sin\left(\theta+\frac{\pi}{6}\right)+Cos\left(\theta+\frac{\pi}{3}\right) = Sin\,\theta\,Cos\,\frac{\pi}{6}+Cos\theta\,Sin\frac{\pi}{6}+Cos\theta\,Cos\frac{\pi}{3}-Sin\theta\,Sin\frac{\pi}{3}$

$= \cancel{Sin\theta\frac{\sqrt{3}}{2}} + Cos\,\theta\,\frac{1}{2} + Cos\,\theta\,\frac{1}{2} - \cancel{Sin\theta\frac{\sqrt{3}}{2}} = Cos\,\theta\left(\frac{1}{2}+\frac{1}{2}\right) = Cos\,\theta\,(1) = Cos\,\theta = R.H.S$

iv. $\mathbf{\frac{Sin\theta-Cos\theta.\tan\theta/2}{Cos\theta+Sin\theta.\tan\theta/2} = \tan\theta/2}$

Sol: L.H.S $= \frac{Sin\theta-Cos\theta.\tan\theta/2}{Cos\theta+Sin\theta.\tan\theta/2}$

$= \frac{Sin\theta-Cos\theta.\frac{Sin\theta/2}{Cos\theta/2}}{Cos\theta+Sin\theta.\frac{Sin\theta/2}{Cos\theta/2}} = \frac{\frac{Sin\theta\,Cos\theta/2-Cos\theta\,Sin\theta/2}{Cos\theta/2}}{\frac{Cos\theta\,Cos\theta+Sin\theta\,Sin\theta/2}{Cos\theta/2}}$

$= \frac{\frac{Sin(\theta-\theta/2)}{Cos\theta/2}}{\frac{Cos(\theta-\theta/2)}{Cos\theta/2}} = \frac{Sin\theta/2}{\cancel{Cos\theta/2}} \times \frac{\cancel{Cos\theta/2}}{Cos\theta/2} = \tan\theta/2 = R.H.S$

v. $\mathbf{\frac{1-\tan\theta.\tan\varphi}{1+\tan\theta.\tan\varphi} = \frac{Cos(\theta+\varphi)}{Cos(\theta-\varphi)}}$

Sol: L.H.S $= \frac{1-\tan\theta.\tan\phi}{1+\tan\theta.\tan\phi} = \frac{1-\frac{Sin\theta}{Cos\theta}\frac{Sin\phi}{Cos\phi}}{1+\frac{Sin\theta}{Cos\theta}\frac{Sin\phi}{Sin\phi}}$

$= \frac{\frac{Cos\theta\,Cos\phi-Sin\theta\,Sin\phi}{Cos\theta\,Cos\phi}}{\frac{Cos\theta\,Cos\phi+Sin\theta\,Sin\phi}{Cos\theta\,Cos\phi}} = \frac{Cos(\theta+\phi)}{Cos(\theta-\phi)} = \frac{Cos(\theta+\varphi)}{\cancel{Cos\theta\,Cos\varphi}} \times \frac{\cancel{Cos\theta\,Cos\varphi}}{Cos(\theta-\varphi)} = \frac{Cos(\theta+\phi)}{Cos(\theta-\phi)}$

Q4. Show that $\mathbf{Cos(\alpha+\beta)\ Cos\ (\alpha-\beta) = Cos^2\alpha - Sin^2\beta = Cos^2\beta - Sin^2\alpha}$

Sol: L.H.S $= Cos(\alpha+\beta).Cos(\alpha-\beta)$

$= (Cos\alpha\ Cos\beta - Sin\alpha\ Sin\beta).(Cos\alpha\ Cos\beta + Sin\alpha\ Sin\beta)$

$= (Cos\alpha\ Cos\beta)^2 - (Sin\alpha\ Sin\beta)^2 = Cos^2\alpha\ Cos^2\beta - Sin^2\alpha\ Sin^2\beta$ _____ I

$= Cos^2\alpha\ (1 - Sin^2\beta) - (1 - Cos^2\alpha)\ Sin^2\beta$

$= Cos^2\alpha - \cancel{Cos^2\alpha Sin^2\beta} - Sin^2\beta + \cancel{Cos^2\alpha Sin^2\beta}$

$= Cos^2\alpha - Sin^2\beta$ ***Result I***

$= (1 - Sin^2\alpha) - (1 - Cos^2\beta)$

$= \cancel{1} - sin^2\alpha - \cancel{1} + cos^2\beta$

$= Cos^2\beta - Sin^2\alpha$ ***Result II***

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Q5. Show that $\dfrac{Sin(\alpha+\beta)+Sin(\alpha-\beta)}{Cos(\alpha+\beta)+Cos(\alpha-\beta)} = \tan\alpha$

Sol: L.H.S $= \dfrac{Sin(\alpha+\beta)+Sin(\alpha-\beta)}{Cos(\alpha+\beta)+Cos(\alpha-\beta)} = \dfrac{Sin\alpha\,Cos\beta + Cos\alpha\,Sin\beta + Sin\alpha\,Cos\beta - Cos\alpha\,Sin\beta}{Cos\alpha\,Cos\beta - Sin\alpha\,Sin\beta + Cos\alpha\,Cos\beta + Sin\alpha\,Sin\beta}$

$= \dfrac{2\,Sin\alpha\,Cos\beta}{2\,Cos\alpha\,Cos\beta} = \tan\alpha$ = R.H.S

Q6(i). Show that $\sin^2\left(\alpha+\frac{\beta}{2}\right)-\sin^2\left(\alpha-\frac{\beta}{2}\right)=\sin 2\alpha.\sin\beta$

Sol: $\text{L.H.S} = \left(\sin\alpha\cos\frac{\beta}{2}+\cos\alpha\sin\frac{\beta}{2}\right)^2-\left(\sin\alpha\cos\frac{\beta}{2}-\cos\alpha\sin\frac{\beta}{2}\right)^2$

$=\left(\sin^2\alpha\cos^2\frac{\beta}{2}+\cos^2\alpha\sin^2\frac{\beta}{2}+2\sin\alpha\cos\frac{\beta}{2}\cos\alpha\sin\frac{\beta}{2}\right)-\left(\sin^2\alpha\cos^2\frac{\beta}{2}+\cos^2\alpha\sin^2\frac{\beta}{2}-2\sin\alpha\cos\frac{\beta}{2}\cos\alpha\sin\frac{\beta}{2}\right)$

$=\sin^2\alpha\cos^2\frac{\beta}{2}+\cos^2\alpha\sin^2\frac{\beta}{2}+2\sin\alpha\cos\frac{\beta}{2}\cos\alpha\sin\frac{\beta}{2}-\sin^2\alpha\cos^2\frac{\beta}{2}-\cos^2\alpha\sin^2\frac{\beta}{2}+2\sin\alpha\cos\frac{\beta}{2}\cos\alpha\sin\frac{\beta}{2}$

$=4\sin\alpha\cos\frac{\beta}{2}\cos\alpha\sin\frac{\beta}{2}=2\sin\alpha\cos\alpha.2\sin\frac{\beta}{2}\cos\frac{\beta}{2}=\sin 2\alpha\sin\beta=R.H.S$

Q6(ii). Show that: $\sin^2\alpha+\sin^2\beta+\cos^2(\alpha+\beta)+2\sin\alpha.\sin\beta.\cos(\alpha+\beta)=1$

Sol: $\text{L.H.S}=\sin^2\alpha+\sin^2\beta+\cos^2(\alpha+\beta)+2\sin\alpha.\sin\beta.\cos(\alpha+\beta)$

$=\sin^2\alpha+\sin^2\beta+\cos(\alpha+\beta)\left[\cos(\alpha+\beta)+2\sin\alpha\sin\beta\right]$

$=\sin^2\alpha+\sin^2\beta+\cos(\alpha+\beta)\left[\cos\alpha\cos\beta-\sin\alpha\sin\beta+2\sin\alpha\sin\beta\right]$

$=\sin^2\alpha+\sin^2\beta+\cos(\alpha+\beta)\left[\cos\alpha\cos\beta+\sin\alpha\sin\beta\right]$

$=\sin^2\alpha+\sin^2\beta+(\cos\alpha\cos\beta-\sin\alpha\sin\beta)(\cos\alpha\cos\beta+\sin\alpha\sin\beta)$

$=\sin^2\alpha+\sin^2\beta+\cos^2\alpha\cos^2\beta-\sin^2\alpha\sin^2\beta$

$=\sin^2\alpha+\sin^2\beta+\cos^2\alpha(1-\sin^2\beta)-(1-\cos^2\alpha)\sin^2\beta$

$=\sin^2\alpha+\sin^2\beta+\cos^2\alpha-\cos^2\alpha\sin^2\beta-\sin^2\beta+\cos^2\alpha\sin^2\beta$

$=\sin^2\alpha+\sin^2\beta+\cos^2\alpha-\sin^2\beta=\sin^2\alpha+\cos^2\alpha+\sin^2\beta-\sin^2\beta$

$=1+0=1$

Q7. Show that:

i. $\cos(\alpha-\beta)=\dfrac{1+\tan\alpha\tan\beta}{\sec\alpha\sec\beta}$

Sol:

R.H.S $= \dfrac{1+\tan\alpha\tan\beta}{\sec\alpha\sec\beta}$

$= \dfrac{1+\frac{\sin\alpha}{\cos\alpha}\cdot\frac{\sin\beta}{\cos\beta}}{\frac{1}{\cos\alpha}\cdot\frac{1}{\cos\beta}} = \dfrac{\frac{\cos\alpha\cos\beta+\sin\alpha\sin\beta}{\cos\alpha\cos\beta}}{\frac{1}{\cos\alpha\cos\beta}} = \dfrac{\cos(\alpha-\beta)}{\cos\alpha\cos\beta}\times\cos\alpha\cos\beta = \cos(\alpha-\beta)=L.H.S$

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ii. $\sin(\alpha+\beta)=\dfrac{1+\cot\alpha\tan\beta}{\operatorname{cosec}\alpha\sec\beta}$

Sol: $\text{R.H.S}=\dfrac{1+\cot\alpha\tan\beta}{\operatorname{cosec}\alpha\sec\beta}$

$$=\frac{1+\dfrac{\cos\alpha}{\sin\alpha}\cdot\dfrac{\sin\beta}{\cos\beta}}{\dfrac{1}{\sin\alpha}\dfrac{1}{\cos\beta}}=\frac{\dfrac{\sin\alpha\cos\beta+cs\alpha\sin\beta}{\cos\alpha\cos\beta}}{\dfrac{1}{\sin\alpha\cos\beta}}$$

$$=\frac{\sin(\alpha+\beta)}{\cancel{\sin\alpha\cos\beta}}\times\cancel{\sin\alpha\cos\beta}=\sin(\alpha+\beta)=L.H.S$$

iii. $\cot(\alpha-\beta)=\dfrac{\cot\alpha\cot\beta+1}{\cot\beta-\cot\alpha}$

Sol: Let $\text{R.H.S}=\dfrac{\cot\alpha\cot\beta+1}{\cot\beta-\cot\alpha}$

$$=\frac{\dfrac{1}{\tan\alpha\tan\beta}+1}{\dfrac{1}{\tan\beta}-\dfrac{1}{\tan\alpha}}=\frac{\dfrac{1+\tan\alpha\tan\beta}{\tan\alpha\tan\beta}}{\dfrac{\tan\alpha-\tan\beta}{1+\tan\alpha\tan\beta}}=\frac{1+\tan\alpha\tan\beta}{\cancel{\tan\alpha\tan\beta}}\times\frac{\cancel{\tan\alpha\tan\beta}}{\tan\alpha-\tan\beta}$$

$$=\frac{1+\tan\alpha\tan\beta}{\tan\alpha-\tan\beta}=\frac{1}{\dfrac{\tan\alpha-\tan\beta}{1+\tan\alpha\tan\beta}}=\frac{1}{\tan(\alpha-\beta)}=\cot(\alpha-\beta)=L.H.S$$

iv. $\dfrac{\tan\alpha+\tan\beta}{\tan\alpha-\tan\beta}=\dfrac{\sin(\alpha+\beta)}{\sin(\alpha-\beta)}$

Sol: $\text{L.H.S}=\dfrac{\tan\alpha+\tan\beta}{\tan\alpha-\tan\beta}=\dfrac{\dfrac{\sin\alpha}{\cos\alpha}+\dfrac{\sin\beta}{\cos\beta}}{\dfrac{\sin\alpha}{\cos\alpha}-\dfrac{\sin\beta}{\cos\beta}}=\dfrac{\dfrac{\sin\alpha\cos\beta+\cos\alpha\sin\beta}{\cos\alpha\cos\beta}}{\dfrac{\sin\alpha\cos\beta-\cos\alpha\sin\beta}{\cos\alpha\cos\beta}}$

$$=\frac{\sin(\alpha+\beta)}{\cancel{\cos\alpha\cos\beta}}\times\frac{\cancel{\cos\alpha\cos\beta}}{\sin(\alpha-\beta)}=\frac{\sin(\alpha+\beta)}{\sin(\alpha-\beta)}=R.H.S$$

v. $\cot(\alpha+\beta)=\dfrac{\cot\alpha\cot\beta-1}{\cot\alpha+\cot\beta}$

Sol: $\text{R.H.S}=\dfrac{\cot\alpha\cot\beta-1}{\cot\alpha+\cot\beta}=\dfrac{\dfrac{1}{\tan\alpha\tan\beta}-1}{\dfrac{1}{\tan\alpha}+\dfrac{1}{\tan\beta}}=\dfrac{\dfrac{1-\tan\alpha\tan\beta}{\tan\alpha\tan\beta}}{\dfrac{\tan\alpha+\tan\beta}{\tan\alpha\tan\beta}}$

$$=\frac{1-\tan\alpha\tan\beta}{\cancel{\tan\alpha\tan\beta}}\times\frac{\cancel{\tan\alpha\tan\beta}}{\tan\alpha+\tan\beta}=\frac{1-\tan\alpha\tan\beta}{\tan\alpha+\tan\beta}=\frac{1}{\dfrac{\tan\alpha+\tan\beta}{1-\tan\alpha\tan\beta}}$$

$$=\frac{1}{\tan(\alpha+\beta)}=\cot(\alpha+\beta)=L.H.S$$

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Q8. If $\sin\alpha = \frac{24}{25}$ and $\cos\beta = \frac{20}{29}$, where $0 < \alpha < \frac{\pi}{2}$ and $0 < \beta < \frac{\pi}{2}$ show that $\sin(\alpha - \beta) = \frac{333}{725}$.

Sol: $\cos^2\alpha = 1 - \sin^2\alpha$

$$= 1 - \left(\frac{24}{25}\right)^2 = 1 - \frac{576}{625} = \frac{625 - 576}{625}$$

$$\cos^2\alpha = \frac{49}{625}$$

$$\Rightarrow \cos\alpha = \pm\frac{7}{25} = \frac{7}{25}(\alpha \text{ in I quad})$$

$$\sin^2\beta = 1 - \cos^2\beta$$

$$= 1 - \left(\frac{20}{29}\right)^2 = 1 - \frac{400}{841} = \frac{841 - 400}{841}$$

$$\sin^2\beta = \frac{441}{841} \Rightarrow \sin\beta = \pm\frac{21}{29} = \frac{21}{29}(\beta \text{ in I quad})$$

Using formula $\sin(\alpha - \beta) = \sin\alpha\cos\beta - \cos\alpha\sin\beta$

$$\sin(\alpha - \beta) = \left(\frac{24}{25}\right)\left(\frac{20}{29}\right) - \left(\frac{7}{25}\right)\left(\frac{21}{29}\right)$$

$$= \frac{480}{725} - \frac{147}{725} = \frac{480 - 147}{725} = \frac{333}{725}$$

Q9. If $\sin\alpha = \frac{-8}{17}$ and $\cos\beta = -\frac{4}{5}$, where $\frac{3\pi}{2} < \alpha < 2\pi$ and $\pi < \beta < \frac{3\pi}{2}$ Find the following.

i. $\sin(\alpha + \beta)$ ii. $\cos(\alpha + \beta)$ iii. $\tan(\alpha + \beta)$

iv. $\sin(\alpha - \beta)$ v. $\cos(\alpha - \beta)$ vi. $\tan(\alpha - \beta)$

In which quadrants do the terminal sides of the angles of measures $(\alpha + \beta)$ and $(\alpha - \beta)$ lie?

Sol: Given that α is in IV-Quad and β is in III-Quad

$\cos^2\alpha = 1 - \sin^2\alpha$

$$= 1 - \left(\frac{-8}{17}\right)^2 = 1 - \frac{64}{289} = \frac{289 - 64}{289}$$

$$\cos^2\alpha = \frac{225}{289} \Rightarrow \cos\alpha = \pm\frac{15}{17}$$

$$\cos\alpha = \frac{15}{17} \qquad (\alpha \text{ in IV quad})$$

$$\tan\alpha = \frac{\sin\alpha}{\cos\alpha} = \frac{-8/17}{15/17} = -\frac{8}{15}$$

$\sin^2\beta = 1 - \cos^2\beta$

$$= 1 - \left(\frac{-4}{5}\right)^2 = 1 - \frac{16}{25} = \frac{9}{25}$$

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$$\sin\beta = \pm\frac{3}{5} = \frac{-3}{5} \qquad (\beta \text{ in III quad})$$

$$\Rightarrow \sin\beta = \frac{-3}{5}$$

$$\tan\beta = \frac{\sin\beta}{\cos\beta} = \frac{\cancel{-3/5}}{\cancel{-4/5}} = \frac{3}{4}$$

i. $\sin(\alpha+\beta)$

Sol: $\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$

$$= \left(\frac{-8}{17}\right)\left(-\frac{4}{5}\right) + \left(\frac{15}{17}\right)\left(\frac{-3}{5}\right) = \frac{32}{85} - \frac{45}{85} = \frac{-13}{85}$$

ii. $\cos(\alpha+\beta)$

Sol: $\cos(\alpha+\beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta$

$$= \left(\frac{15}{17}\right)\left(\frac{-4}{5}\right) - \left(\frac{-8}{17}\right)\left(\frac{-3}{5}\right) = \frac{-60}{85} - \frac{24}{85} = \frac{-84}{85}$$

iii. $\tan(\alpha+\beta)$

Sol: $$\tan(\alpha+\beta) = \frac{\tan\alpha+\tan\beta}{1-\tan\alpha\tan\beta} = \frac{\frac{-8}{15}+\frac{3}{4}}{1-\left(\frac{-8}{15}\right)\left(\frac{3}{4}\right)} = \frac{\frac{-8}{15}+\frac{3}{4}}{1+\frac{24}{60}} = \frac{\frac{-32+45}{\cancel{60}}}{\frac{60+24}{\cancel{60}}} = \frac{13}{84}$$

iv. $\sin(\alpha-\beta)$

Sol: $\sin(\alpha-\beta) = \sin\alpha\cos\beta - \cos\alpha\sin\beta$

$$= \left(\frac{-8}{17}\right)\left(-\frac{4}{5}\right) - \left(\frac{15}{17}\right)\left(\frac{-3}{5}\right) = \frac{32}{85} + \frac{45}{85} = \frac{77}{85}$$

v. $\cos(\alpha-\beta)$

Sol: $\cos(\alpha-\beta) = \cos\alpha\cos\beta + \sin\alpha\sin\beta$

$$= \left(\frac{15}{17}\right)\left(\frac{-4}{5}\right) + \left(\frac{-8}{17}\right)\left(\frac{-3}{5}\right) = \frac{-60}{85} + \frac{24}{85} = \frac{-36}{85}$$

vi. $\tan(\alpha-\beta)$

Sol: $$\tan(\alpha-\beta) = \frac{\tan\alpha-\tan\beta}{1+\tan\alpha\tan\beta} = \frac{\frac{-8}{15}-\frac{3}{4}}{1+\left(\frac{-8}{15}\right)\left(\frac{3}{4}\right)} = \frac{\frac{-32-45}{\cancel{60}}}{\frac{60-24}{\cancel{60}}} = \frac{-77}{36}$$

$\sin(\alpha+\beta) = -ve$, $\cos(\alpha+\beta) = -ve$

but $\tan(\alpha+\beta) = +ve$ so $\alpha+\beta$ is in III quad

Similarly,

$\sin(\alpha-\beta) = +ve$, $\cos(\alpha-\beta) = -ve$, $\tan(\alpha-\beta) = -ve$

So $\alpha-\beta$ is in II quad

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10.(i) Find $\sin(\alpha+\beta)$ and $\cos(\alpha+\beta)$ given that $\tan\alpha = \frac{3}{4}$, $\cos\beta = \frac{5}{13}$ and neither the terminal side of the angle of measure α nor that β is in the I quadrant.

Given $\tan\alpha = +ve$ but not in I so α is in III quad
Similarly, $\cos\beta = +ve$ but β not is in I so β is in IV quad

Sol: $1+\tan^2\alpha = \sec^2\alpha \Rightarrow 1+\frac{9}{16} = \sec^2\alpha = \frac{25}{16} \Rightarrow \sec\alpha = \pm\frac{5}{4}$

$\Rightarrow \cos\alpha = \pm\frac{4}{5} \Rightarrow \cos\alpha = \frac{-4}{5}$ (Because α in III)

$\sin^2\alpha = 1-\cos^2\alpha = 1-\left(\frac{-4}{5}\right)^2 = 1-\frac{16}{25} = \frac{9}{25} \Rightarrow \sin\alpha = \pm\frac{3}{5}$

$\sin\alpha = -\frac{3}{5}$ (Because α is in III)

$\sin^2\beta = 1-\cos^2\beta = 1-\left(\frac{5}{13}\right)^2 = 1-\frac{25}{169} = \frac{169-25}{169} = \frac{144}{169} = \sin\beta = \pm\frac{12}{13}$

$\sin\beta = \frac{-12}{13}$ (Because β is in IV)

a. $\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$

Sol. $= \left(\frac{-3}{5}\right)\left(\frac{5}{13}\right)+\left(\frac{-4}{5}\right)\left(\frac{-12}{13}\right) = \frac{-15}{65}+\frac{48}{65} = \frac{-15+48}{65} = \frac{33}{65}$

b. $\cos(\alpha+\beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta$

Sol. $= \left(\frac{-4}{5}\right)\left(\frac{5}{13}\right)-\left(\frac{-3}{5}\right)\left(\frac{-12}{13}\right) = \frac{-20}{65}-\frac{36}{65} = \frac{-20-36}{65} = \frac{-56}{65}$

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10 (ii) $\tan\alpha = -\frac{15}{8}$, $\sin\beta = -\frac{7}{25}$ and neither the terminal side of the angle of measure α nor that β is in the IV quadrant.

Sol: $1+\tan^2\alpha = \sec^2\alpha \Rightarrow 1+\left(\frac{-15}{8}\right)^2 = \sec^2\alpha$

Given α not in IV and $\tan\alpha = -ve$ so α in II Similarly β in III

$1+\frac{225}{16} = Sec^2\alpha \Rightarrow \frac{289}{64} = Sec^2\alpha \Rightarrow \sec\alpha = \pm\frac{17}{8} \Rightarrow Cos\alpha = \pm\frac{8}{17}$

$\boxed{\cos\alpha = -8/17}$ (Because α is in II)

$\sin^2\alpha = 1-\cos^2\alpha = 1-\left(\frac{-8}{17}\right)^2 = 1-\frac{64}{289} = \frac{225}{289} \Rightarrow Sin\alpha = \pm\frac{15}{17}$

$\boxed{\sin\alpha = \frac{15}{17}}$ (Because α is in II)

$\cos^2\beta = 1-\sin^2\beta = 1-\left(\frac{-7}{25}\right)^2 = 1-\frac{49}{625} = \frac{625-64}{625} = \frac{576}{625}$

$\cos\beta = \pm\frac{24}{25} \Rightarrow \boxed{\cos\beta = -\frac{24}{25}}$ (Because β is in III)

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Now

a. $\text{Sin}(\alpha+\beta) = \text{Sin}\,\alpha\ \text{Cos}\,\beta + \text{Cos}\,\alpha\ \text{Sin}\,\beta$

Sol: $=\left(\frac{15}{17}\right)\left(\frac{-24}{25}\right)+\left(\frac{-8}{17}\right)\left(\frac{-7}{25}\right)=\frac{-360}{425}+\frac{56}{425}=\frac{-360+56}{425}=\frac{-304}{425}$

b. $\text{Cos}(\alpha+\beta) = \text{Cos}\,\alpha\ \text{Cos}\,\beta - \text{Sin}\,\alpha\ \text{Sin}\,\beta = \left(\frac{-8}{17}\right)\left(\frac{-24}{25}\right)-\left(\frac{15}{17}\right)\left(\frac{-7}{25}\right)$

Sol: $=\frac{192}{425}+\frac{105}{425}=\frac{192+105}{425}=\frac{297}{425}$

Q11. **Prove that:** $\frac{\cos 19^0+\sin 19^0}{\cos 19^0-\sin 19^0}=\tan 64^0$

Sol: $\text{R.H.S} = \tan\left(64^0\right)=\tan\left(45^0+19^0\right)$

$$=\frac{\tan 45^0+\tan 19^0}{1-\tan 45^0\tan 19^0}=\frac{1+\tan 19^0}{1-\tan 19^0}=\frac{1+\frac{\sin 19^0}{\cos 19^0}}{1-\frac{\sin 19^0}{\cos 19^0}}=\frac{\frac{\cos 19^0+\sin 19^0}{\cancel{\cos 19^0}}}{\frac{\cos 19^0-\sin 19^0}{\cancel{\cos 19^0}}}=\frac{\cos 19^0+\sin 19^0}{\cos 19^0-\sin 19^0}=L.H.S$$

Q12. **Prove that** $\cos\left(60^0+\theta\right)\cos\left(60^0-\theta\right)+\sin\left(60^0+\theta\right)\sin\left(60^0-\theta\right)=\cos 2\theta$

Sol: $\text{L.H.S} = \cos\left(60^0+\theta\right)\cos\left(60^0-\theta\right)+\sin\left(60^0+\theta\right)\sin\left(60^0-\theta\right)$

Using formula: $\cos\left(\alpha-\beta\right)=\cos\alpha\cos\beta+\sin\alpha\sin\beta$

$$=\cos\left[\left(60^0+\theta\right)-\left(60^0-\theta\right)\right]=\cos\left(\cancel{60^0}+\theta-\cancel{60^0}+\theta\right)=\cos 2\theta=R.H.S$$

Q13. **If α , β , γ are angles of triangle ABC, Show that Cot α /2 + Cot β /2 + Cot γ /2 = Cot α /2Cot β /2Cot γ /2**

Sol: We know that $\alpha + \beta + \gamma = 180° \Rightarrow \alpha + \beta = 180° - \gamma$

Divide both side by '2' $\frac{\alpha+\beta}{2}=\frac{180^o-\gamma}{2}\Rightarrow\frac{\alpha+\beta}{2}=90^o-\frac{\gamma}{2}\Rightarrow\frac{\alpha}{2}+\frac{\beta}{2}=90^o-\frac{\gamma}{2}$

$\tan(\alpha/2 + \beta/2) = \tan(90° - \gamma/2) \Rightarrow \frac{\tan\alpha/2+\tan\beta/2}{1-\tan\alpha/2\tan\beta/2} = \text{Cot}\,\gamma/2$

$$\frac{\frac{1}{Cot\alpha/2}+\frac{1}{Cot\beta/2}}{1-\frac{1}{Cot\alpha/2Cot\beta/2}} = \text{Cot}\,\gamma/2 \Rightarrow \frac{\frac{Cot\frac{\alpha}{2}+Cot\frac{\beta}{2}}{Cot\frac{\alpha}{2}Cot\frac{\beta}{2}}}{\frac{Cot\frac{\alpha}{2}Cot\frac{\beta}{2}-1}{Cot\frac{\alpha}{2}Cot\frac{\beta}{2}}} = \text{Cot}\,\frac{\gamma}{2}$$

$$\frac{Cot\alpha/2+Cot\beta/2}{\cancel{Cot\alpha/2Cot\beta/2}}\times\frac{\cancel{Cot\alpha/2Cot\beta/2}}{Cot\alpha/2\,Cot\beta/2-1} = \text{Cot}\,\gamma/2$$

$\text{Cot}\,\alpha/2 + \text{Cot}\,\beta/2 = \text{Cot}\,\gamma/2\,(\text{Cot}\,\alpha/2\text{Cot}\,\beta/2 - 1)$

$\text{Cot}\,\alpha/2 + \text{Cot}\,\beta/2 = \text{Cot}\,\alpha/2\text{Cot}\,\beta/2\text{Cot}\,\gamma/2 - \text{Cot}\,\gamma/2$

$\text{Cot}\,\alpha/2 + \text{Cot}\,\beta/2 + \text{Cot}\,\gamma/2 = \text{Cot}\,\alpha/2\text{Cot}\,\beta/2\text{Cot}\,\gamma/2$

Hence proved.

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14. If $\alpha+\beta+\gamma=180^{\circ}$, Show that $\cot\alpha\ \cot\beta+\cot\beta\ \cot\gamma+\cot\gamma\ \cot\alpha=1$

Sol: Given $\alpha+\beta+\gamma=180^{\circ}\Rightarrow\alpha+\beta=180^{\circ}-\gamma\Rightarrow\tan(\alpha+\beta)=\tan(180^{\circ}-\gamma)$

$$\frac{\tan\alpha+\tan\beta}{1-\tan\alpha\tan\beta}=-\tan\gamma\Rightarrow\frac{\frac{1}{Cot\alpha}+\frac{1}{Cot\beta}}{1-\frac{1}{Cot\alpha Cot\beta}}=-\frac{1}{Cot\gamma}$$

$$\frac{\frac{Cot\alpha+Cot\beta}{Cot\alpha Cot\beta}}{\frac{Cot\alpha Cot\beta-1}{Cot\alpha Cot\beta}}=-\frac{1}{Cot\gamma}\Rightarrow\frac{Cot\alpha+Cot\beta}{\cancel{Cot\alpha Cot\beta}}\times\frac{\cancel{Cot\alpha Cot\beta}}{Cot\alpha\,Cot\beta-1}=-\frac{1}{Cot\gamma}$$

$$(Cot\alpha+Cot\beta)(Cot\gamma)=-(Cot\alpha\,Cot\beta-1)$$

$$Cot\alpha\,Cot\gamma+Cot\beta Cot\gamma=-Cot\alpha Cot\beta+1$$

$$Cot\alpha\,Cot\beta+Cot\beta\,Cot\gamma+Cot\gamma\,Cot\alpha=1$$

Hence Proved.

15. **Express the following in the form $r\sin(\theta+\phi)$ or $r\sin(\theta-\phi)$ where terminal sides of the angles of measures θ and ϕ are in the first quadrant:**

i. $24\sin\theta+7\cos\theta$

Sol: Take

$24=r\cos\phi$ ____ I & $7=r\sin\phi$ _____ II

Squaring and Adding (I) and (II)

As $r^2\cos^2\phi+r^2\sin^2\phi=(24)^2+(7)^2$

$r^2(\cos^2\phi+\sin^2\phi)=576+49$

$r^2=625\Rightarrow r=25$

Dividing (II) By (I)

$$\frac{r\sin\phi}{r\cos\phi}=\frac{7}{24}\Rightarrow\tan\phi=\frac{7}{24}$$

Then $24\sin\theta+7\cos\theta=r\cos\phi\sin\theta+r\sin\phi\cos\theta$

$=r(\sin\theta\cos\phi+\cos\theta\sin\phi)=r\sin(\theta+\phi)=25\sin(\theta+\phi)$

Where $r=25$ and $\tan\phi=\frac{7}{24}$

ii. $12\sin\theta-5\cos\theta$

Sol: Take $12=r\cos\phi$ ____ (I)

and $5=r\sin\phi$ ____ (II)

Squaring and adding (1) and (2)

$(12)^2+(5)^2=r^2\cos\phi+r^2\sin^2\phi c$

$144+25=r^2(\cos^2\phi+\sin^2\phi)$

$169=r^2(1)\Rightarrow r=13$

Divide (II) by (I)

$$\frac{r\sin\phi}{r\cos\phi}=\frac{5}{12}$$

$$\tan\phi=\frac{5}{12}$$

Then

$r\cos\phi\sin\theta-r\sin\phi\cos\theta$

$=r(\sin\theta\cos\phi-\cos\theta\sin\phi)=r\sin(\theta-\phi)$

$=13\sin(\theta-\phi)$

Where $r=13$ and $\tan\phi=\frac{5}{12}$

iii. $\sin\theta-\cos\theta$

Sol: Take $1=r\cos\phi$ ___ I & $1=r\sin\phi$ ___ II

By squaring and Adding I & II

$(1)^2+(1)^2=r^2\cos^2\phi+r^2\sin^2\phi$

$1+1=r^2(\cos^2\phi+\sin^2\phi)\Rightarrow 2=r^2\Rightarrow r=\sqrt{2}$

By dividing II by I

$$\frac{\cancel{r}\sin\phi}{\cancel{r}\cos\phi}=\frac{1}{1}\Rightarrow\tan\phi=1$$

$\sin\theta-\cos\theta=r\cos\phi\sin\theta-r\sin\phi\cos\theta$

$=r(\sin\theta\cos\phi-\sin\phi\cos\theta)$

$=r\sin(\theta-\phi)=\sqrt{2}\sin(\theta-\phi)$

Where $r=\sqrt{2}$ *and* $\tan\phi=1$

iv. $8\sin\theta-6\cos\theta$

Sol: Put $8=r\cos\phi$ ___ I & $6=r\sin\phi$ ___ II

Squaring and adding I & II

$r^2\cos^2\phi+r^2\sin^2\phi=(8)^2+(6)^2$

$r^2(\cos^2\phi+\sin^2\phi)=64+36$

$r^2=100\Rightarrow r=10$

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By dividing II by I

$$\frac{r\sin\phi}{r\cos\phi}=\frac{6}{8}\Rightarrow\tan\phi=\frac{3}{4}$$

Then $8\sin\theta-6\cos\theta=r\cos\phi\sin\theta-r\sin\phi\cos\theta$

$=r\left(\sin\theta\cos\phi-\cos\theta\sin\phi\right)$

$=r\sin\left(\theta-\phi\right)=10\sin\left(\theta-\phi\right)$

Where $r=10$ and $\tan\phi=\frac{3}{4}$

v. $\frac{1}{2}\sin\theta+\frac{\sqrt{3}}{2}\cos\theta$

Sol: Put $\frac{1}{2}=r\cos\phi___(I)$

and $\frac{\sqrt{3}}{2}=r\sin\phi___(II)$

Squaring and adding (I) & (II)

$$\left(\frac{1}{2}\right)^2+\left(\frac{\sqrt{3}}{2}\right)^2=r^2\cos^2\phi+r^2\sin^2\phi$$

Divide (II) by (I)

$$\frac{\frac{\sqrt{3}}{2}}{\frac{1}{2}}=\frac{r\sin\phi}{r\cos\phi}$$

$\tan\phi=\sqrt{3}$

$$\frac{1}{4}+\frac{3}{4}=r^2\left(\cos^2\phi+\sin^2\phi\right)$$

$$\frac{4}{4}=r^2\Rightarrow r=1$$

$$\frac{1}{2}\sin\theta+\frac{\sqrt{3}}{2}=r\cos\phi\sin\theta+r\sin\phi\cos\theta$$

$=r\left(\sin\theta\cos\phi+\cos\theta\sin\phi\right)$

$=r\sin\left(\theta+\phi\right)=\sin\left(\theta+\phi\right)$

Where $\tan\phi=\sqrt{3}$ and $r=1$

vi. $13\sin\theta-84\cos\theta$

Sol: Put

$13=r\cos\phi___I,\quad 84=r\sin\phi___II$

Squaring and Adding I & II

$$r^2\cos\phi+r^2\sin^2\phi=(13)^2+(84)^2$$

$$r^2\left(\cos\phi+\sin^2\phi\right)=169+7056$$

$$r^2=7225\Rightarrow r=85$$

By dividing II by I

$$\frac{r\sin\phi}{r\cos\phi}=\frac{84}{13}\Rightarrow\tan\phi=\frac{84}{13}$$

Then

$13\sin\theta-84\cos\theta=r\cos\phi\sin\theta-r\sin\phi\cos\theta$

$=r\left(\sin\theta\cos\phi-\cos\theta\sin\phi\right)$

$=r\sin\left(\theta-\phi\right)=85\sin\left(\theta-\phi\right)$

Where $r=85$ and $\tan\phi=\frac{84}{13}$

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